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Project Title:	Sub-regional Assessment of Bone Loss in the Axial Skeleton in Long-Term Spaceflight		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Leblanc, Adrian (Baylor College of Medicine) Genant, Harry (University of California) Lu, Ying (San Francisco)		
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Task Description:	Bone loss is an established risk factor for long-duration space flight. Experience with the Mir missions and long-term bedrest studies has demonstrated the regional extent of this bone loss, its large variability among individuals, and the fact that it persists despite countermeasures which preserve skeletal loading. Despite this important initial progress, several key gaps in our knowledge remain. There is little information about (1) the distribution of this loss between the trabecular and cortical envelopes of the central skeleton, (2) the cortical/trabecular distribution of the recovery of bone mass post-flight, and (3) the extent to which these changes compare to the spread of bone mineral density (BMD) measures in healthy men and women. The goal of this scientific proposal is to address these three key issues. We will achieve these goals by carrying out a longitudinal study of 10-15 individuals making 4-6 month flights on the International Space Station (ISS). We are imaging these subjects pre-flight, immediately postflight, and 1-year postflight using Quantitative Computed Tomography (QCT) of the spine and hip, dual x-ray absorptiometry (DXA) of the spine, hip and calcaneus, and calcaneal ultrasound. We are assessing sub-regional changes in the spine and the femur by analyzing the QCT images using newly-developed algorithms which compute BMD in trabecular and cortical sub-regions in the spine and hip. We will compute bone loss and recovery in these sub-regions and compare the measurements to the corresponding DXA measures which integrate larger volumes of trabecular and cortical bone. This comparison will establish the extent to which DXA measurements underestimate significant local changes which may impact fracture risk. To compare these changes to the variation of bone density in healthy normal populations, we have acquired a reference database consisting of proximal femoral and spinal QCT scans of healthy Caucasian-American males and females between the ages of 35-45, and are in the process of acquiring a comparable reference data set for African-Americans. Finally, in order to establish the utility of calcaneal ultrasound as a surrogate for calcaneal x-ray absorptiometry, we will compare calcaneal ultrasound and DXA measurements. To characterize the compartmental bone changes in biomechanically relevant anatomic sub-regions of the spine and proximal femur associated with long-duration exposure to microgravity and with re-adaptation to earth gravity. To compare these changes with those quantified by DXA images. To characterize the changes in quantitative ultrasound parameters (broadband ultrasound attenuation and speed of sound) of the heel and compare these to changes measured by DXA of the heel. By quantifying the extent of bone loss during the mission and the extent to which it is recovered after the mission, this protocol yields information on the extent to which individuals who have undergone long-duration spaceflight may be subject to long-term increased skeletal frailty. By defining the time course of skeletal recovery, it will yield information on the frequency with which crewmembers may participate in long-duration missions. By examining differences between integral bone measures such as DXA and sub-regional measurements it has impact on the meaning of the DXA information in terms of tracking bone loss.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary goal of our study is to determine the extent to which the skeletal changes occurring in long-duration human spaceflight are concentrated in small sub-volumes of bone or are more generally distributed throughout the bone. If bone loss is concentrated locally, then measures of bone density (such as DXA, the standard clinical measurement method) which integrate across large volumes of bone may not adequately assess the change in bone strength which occurs in long-duration spaceflight and in the recovery period after the flight. The primary benefits of our ground study derive from the acquisition of a normative database of spinal and femoral QCT scans of healthy Caucasian- and African American males and females between the ages of 35-45. First, by providing reference bone mineral density values against which patient measurements may be compared, this experiment will help facilitate the application of QCT of the hip to diagnosis of osteoporosis in the elderly population. With the increasing availability of new CT scanners in outpatient clinics and small hospitals, management of osteoporosis in the growing elderly population may become an important application of these devices. Additionally, the acquisition of reference data for Caucasian-American males and for African-Americans of both genders, will benefit groups which have not been the main focus of osteoporosis research efforts, which have traditionally centered on Caucasian females. A second benefit is that the QCT reference data comparing African- and Caucasian-Americans will provide important data on gender and racial differences in bone density and geometry in the hip and spine. Almost all of the studies in this area have utilized DXA, which measures an areal density depending on both on the mineral content and the size of the bone. Thus, if two racial groups have different bone sizes, DXA may provide incorrect estimates of the racial difference in bone density. By utilizing QCT, on the other hand, we will be able to measure the volumetric bone mineral density in the hip and spine as well as the three-dimensional geometric properties of these bones.
	Aim 1 and Aim 2: The initial bone loss study was published in the June 2004 issue of the Journal of Bone and Mineral Research. Pre- and postflight comparison analyses were carried out in 18 subjects. All of these data have been forwarded to the LSDA (Life Sciences Data Archive), and the mean results are extremely close to those reported in 2004. These data indicated that integral bone density and mass were lost at 1-1.5% month, and that trabecular BMD in the proximal femur was lost at a rate 50-100% higher, a difference that was statistically significant even in this small number of subject.s. Thus, these data indicated that the rate of loss of trabecular bone at the hip exceeds that of corresponding integral measures, which are dominated by the relatively slowly changing cortical compartment. They also indicate that there is no specific accelerated loss of bone at the posterior elements. Aim 3 This portion of the study was published in the August 2006 issue of the Journal of Bone and Mineral Research. In the 16 subjects, changes measured between preflight, postflight and one-year were assessed. In the year postflight, we observed that bone mass was recovered after spaceflight, and there was no significant difference at one year compared to preflight. Volumetric bone density, on the other hand, was not recovered after spaceflight. The discrepancy between bone mass and volumetric BMD changes was explained by increases in bone volume in the year after spaceflight. Extension Study Results: Our analyses of the 8 subjects were presented at several meetings: including the 2007 and 2008 HRP Investigators Meetings and the 16th IAAA Humans in Space Symposium held in Beijing May 2007. In the long term results, integral BMD showed changes consistent with an asympotic recovery to preflight bone density. However, this long term pattern in integral bone seemed to conceal disparate behavior of the trabecular and cortical

Task Progress:	bone. This discrepancy was evident for both the spine and the proximal femur. The trabecular bone density tended to show a long term decrease, whereas the cortical bone density increased with time. This may be consistent with an increased role for the cortical compartment in load bearing, when the ability of trabecular bone to recover its mass is hindered by the loss of complete trabecular elements, which can act as a template. Aim 4: We have carried out a study which was presented in poster form at the annual meeting of the American Society of Bone and Mineral Research. This study summarizes the variations in bone density and content found for our young normal reference population. The effect of race and gender on proximal femoral bone mineral density (BMD) and geometry is relevant to understanding racial differences in hip fracture rates. To address this issue, we examined differences in areal BMD, volumetric BMD and femoral neck cross-sectional area (CSA) measured by quantitative computed tomography (QCT), in a biracial cohort of 30 Caucasian-American (CA) men, 25 African-American (AA), 30 CA women and 17 AA women. All subjects were aged 35-45 years and had no disorders affecting bone metabolism. CT (GE-9800Q, 80 kVp, 3-mm slice thickness) images were acquired of the proximal femur. In the QCT images, we quantified femoral neck integral (viBMD), trabecular (vtBMD) volumetric BMD and cortical BMD (vcBMD). To quantify bone size, we measured the femoral neck CSA. A DXA-like areal BMD (aBMD) (g/cm2) was quantified in the total femur region. The statistical significance of race and gender differences was ascertained using a multi-variate regression model age adjusted for age, height and weight. The gender and race sub-groups were well matched in age. Men were taller and heavier than women, but there were no racial differences. AA subjects had larger BMD than CA subjects but equivalent bone size. Men and women were similar in all measures of BMD but men had larger bone size. This data acquisition supported a paper published in 2006 by Meta et al in Bone in which young normal white women were compared to white women between the ages of 70-80 years using the QCT and DXA protocol used to analyze the astronaut data. This is summarized in the abstract below, taken from the paper: Fragility fractures at the trochanter (TR) and the femoral neck (FN) have distinct etiologies, but the underlying age-related structural changes at these proximal femoral sub-regions are poorly understood. 28 young (41 ± 3 years) and 124 elderly (74 ± 3 years) healthy Caucasian women underwent volumetric quantitative computed tomography at the hip. Integral (i), cortical (c) and trabecular (t) bone mineral density and content (BMD, BMC) were measured. Geometric parameters included cross sectional area (CSA), and volumes of the integral, cortical and trabecular regions (VOL). Structural measures included indices of compressive (Compstr) and bending (BSI) strength. After adjusting for height and weight, an F-test was used to compare the TR and the FN mean values between young and elderly and to test for interaction to compare logarithmic difference of young and elderly (log(Young)-log(Elderly), Y/Ed) between the FN and the TR in an ANOCOVA model. All BMC, iBMD and tBMD values were significantly lower in elderly than in young women, with the largest Y/Ed in the FN tBMC and tBMD ($P < 0.0011$ and $P < 0.0001$). cBMD in young and elderly groups was not significantly different at the TR while at the FN it was greater ($P = 0.0075$) in elderlythan young women, showing significant Y/Ed ($P = 0.0003$) dependence on skeletal site. Elderly women had significantly larger iVOL and CSA values ($0.0001< P < 0.0051$), except for the FN iVOL. cVOL values were smaller in elderly than young women ($P < 0.0001$). Y/Ed in bone geometry differed by sub-region only for cVOL measures ($P = 0.0267$). Despite larger CSA and iVOL measures in elderly, the younger women had greater Compstr ($P < 0.0001$) and BSI ($P = 0.0051$). Thus, although both the TR and the FN appear to increase in size with age, this enlargement is insufficient to protect against loss of bone strength. Aim 5: No QUS parameter changed significantly over the mission, a finding inconsistent with a previously published bedrest study. Changes in calcaneal QUS were not correlated with those of hip, spinal or calcaneal DXA. Our results do not support the use of heel measurements, particularly calcaneal QUS, to act as surrogates for central bone mass in long-duration spaceflight. As measured by DXA, the amount of bone loss from the heel was strikingly smaller (0.4 %/month vs 0.8-1.5 %/month) than that observed from the central skeleton. The lower responsiveness of the calcaneal DXA measurements compared to the central DXA measurements are consistent with a previous report, and may reflect the fact that the heel is subject to ground reaction forces during in-flight treadmill exercises which are highly attenuated at the central skeletal sites such as the spine and hip. Interestingly, the rate of bone loss from the heel in this study tended to be smaller than the loss observed in bedrest studies, where treadmill countermeasures have not been employed. The poor responsiveness of QUS in spaceflight is at variance with results of ground-based studies, and may reflect changes in the calcaneal soft tissue caused by sustained weightlessness.

Bibliography Type:	Description: (Last Updated: 03/20/2017)
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